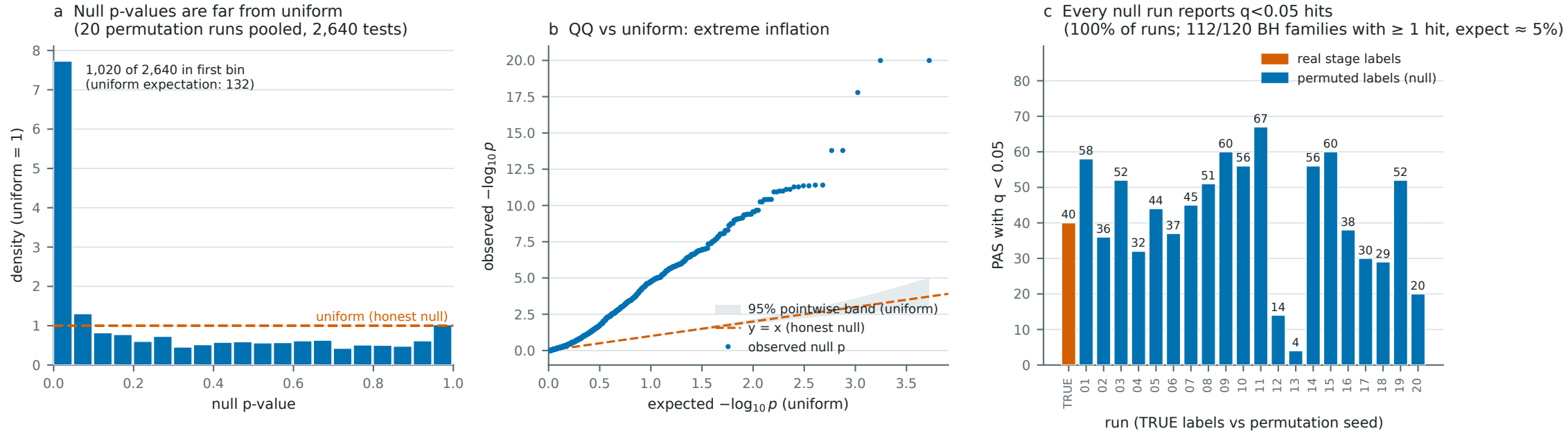


ema switch diff (fisher) is NOT FDR-calibrated under a stage-label permutation null -- Laughney TREG, 3,790 cells, 6 stage pairs per run



Null = obs['stage'] shuffled across cells (label counts kept), full pipeline re-run per permutation, seeds 1-20. 20/20 permutations available when drawn. 39% of null p-values < 0.05 (expect 5%). Drivers of inflation: Fisher on read counts pseudoreplicates (reads within a cell are correlated; documented anti-conservative in ema code), and the top-200-per-cluster marker pre-selection uses the same labels that are then tested (selection bias is deliberately included in this null). Fisher-exact p-values are discrete, so mild deviation from uniform is expected at low counts -- not 40+ orders of magnitude.